

4

Applications of Differentiation



4.1

Maximum and Minimum Values

Maximum and Minimum Values (1 of 1)

Some of the most important applications of differential calculus are *optimization problems*, in which we are required to find the optimal (best) way of doing something.

These problems can be reduced to finding the maximum or minimum values of a function. Let's first explain exactly what we mean by maximum and minimum values.



Absolute and Local Extreme Values

Absolute and Local Extreme Values (1 of 11)

We see that the highest point on the graph of the function f shown in Figure 1 is the point $(3, 5)$.

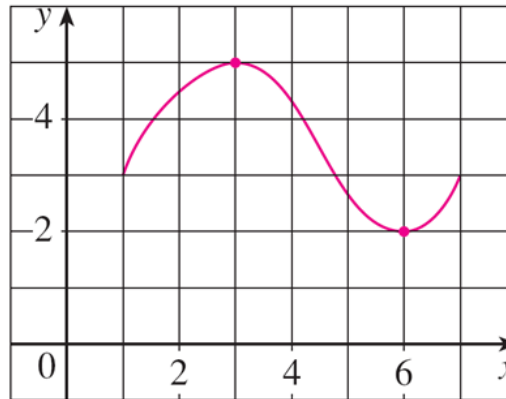


Figure 1

In other words, the largest value of f is $f(3) = 5$. Likewise, the smallest value is $f(6) = 2$.

Absolute and Local Extreme Values (2 of 11)

We say that $f(3) = 5$ is the *absolute maximum* of f and $f(6) = 2$ is the *absolute minimum*. In general, we use the following definition.

1 Definition Let c be a number in the domain D of a function f . Then $f(c)$ is the

- **absolute maximum** value of f on D if $f(c) \geq f(x)$ for all x in D .
- **absolute minimum** value of f on D if $f(c) \leq f(x)$ for all x in D .

global

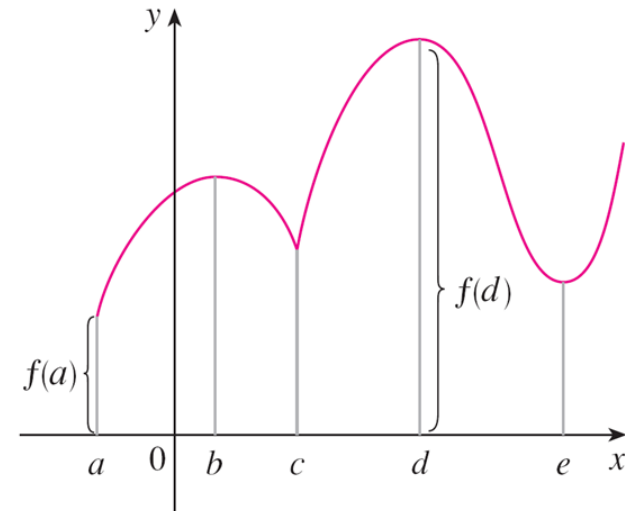
An absolute maximum or minimum is sometimes called a **global** maximum or minimum. The maximum and minimum values of f are called **extreme values** of f .

Absolute and Local Extreme Values (3 of 11)

Figure 2 shows the graph of a function f with absolute maximum at d and absolute minimum at a .

Note that $(d, f(d))$ is the highest point on the graph and $(a, f(a))$ is the lowest point.

In Figure 2, if we consider only values of x near b [for instance, if we restrict our attention to the interval $(a, c]$], then $f(b)$ is the largest of those values of $f(x)$ and is called a *local maximum value* of f .



Abs min $f(a)$, abs max $f(d)$, loc min $f(c)$,
 $f(e)$, loc max $f(b)$, $f(d)$

Figure 2

Absolute and Local Extreme Values (4 of 11)

Likewise, $f(c)$ is called a *local minimum value* of f because $f(c) \leq f(x)$ for x near c [in the interval (b, d) , for instance].

The function f also has a local minimum at e . In general, we have the following definition.

2 Definition The number $f(c)$ is a

- **local maximum** value of f if $f(c) \geq f(x)$ when x is near c .
- **local minimum** value of f if $f(c) \leq f(x)$ when x is near c .

In Definition 2 (and elsewhere), if we say that something is true **near** c , we mean that it is true on some open interval containing c .

Absolute and Local Extreme Values (5 of 12)

For instance, in Figure 3 we see that $f(4) = 5$ is a local minimum because it's the smallest value of f on the interval I .

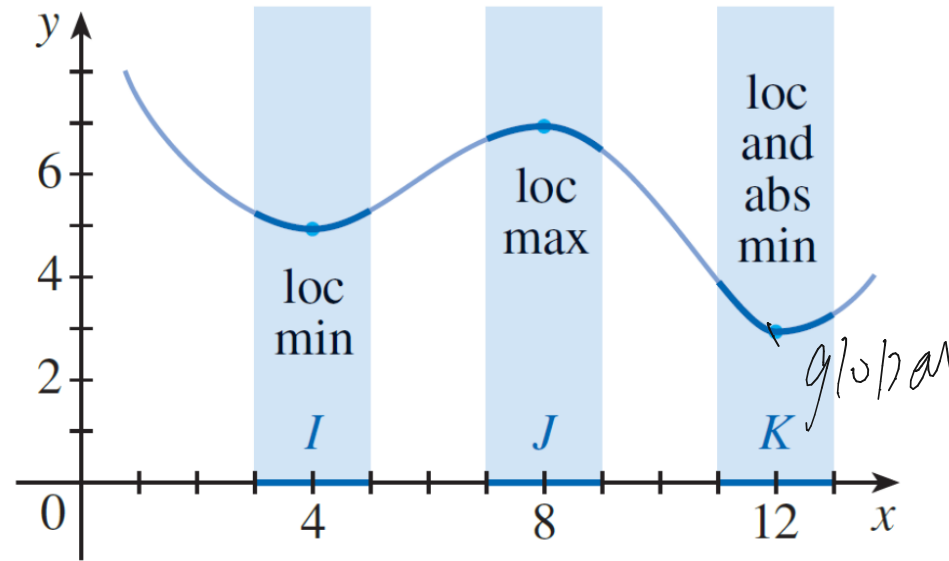


Figure 3

Absolute and Local Extreme Values (6 of 12)

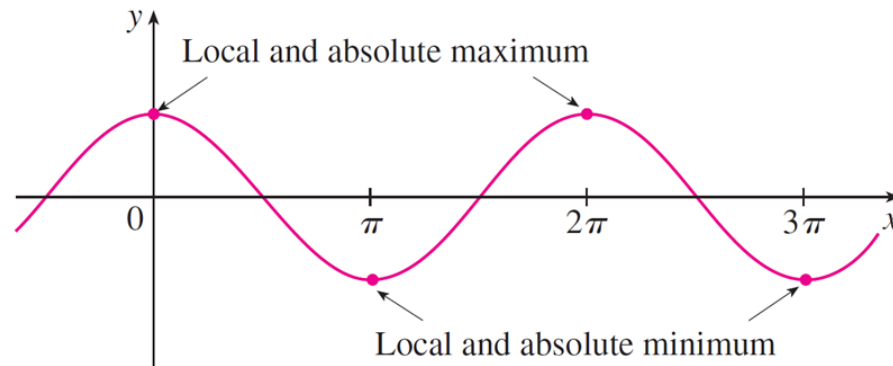
It's not the absolute minimum because $f(x)$ takes smaller values when x is near 12 (in the interval K , for instance).

In fact $f(12) = 3$ is both a local minimum and the absolute minimum.

Similarly, $f(8) = 7$ is a local maximum, but not the absolute maximum because f takes larger values near 1.

Example 2

The function $f(x) = \cos x$ takes on its (local and absolute) maximum value of 1 infinitely many times, because $\cos 2n\pi = 1$ for any integer n and $-1 \leq \cos x \leq 1$ for all x . (See Figure 5.)



$$y = \cos x$$

Figure 5

Likewise, $\cos(2n + 1)\pi = -1$ is its minimum value, where n is any integer.

Absolute and Local Extreme Values (7 of 11)

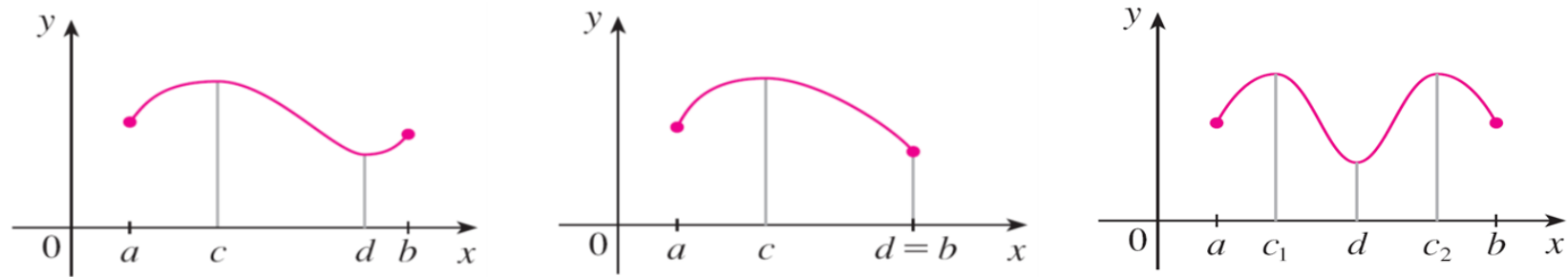
The following theorem gives conditions under which a function is guaranteed to possess extreme values.

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3 The Extreme Value Theorem If f is continuous on a closed interval $[a, b]$, then f attains an absolute maximum value $f(c)$ and an absolute minimum value $f(d)$ at some numbers c and d in $[a, b]$.

Absolute and Local Extreme Values (8 of 11)

The Extreme Value Theorem is illustrated in Figure 8.



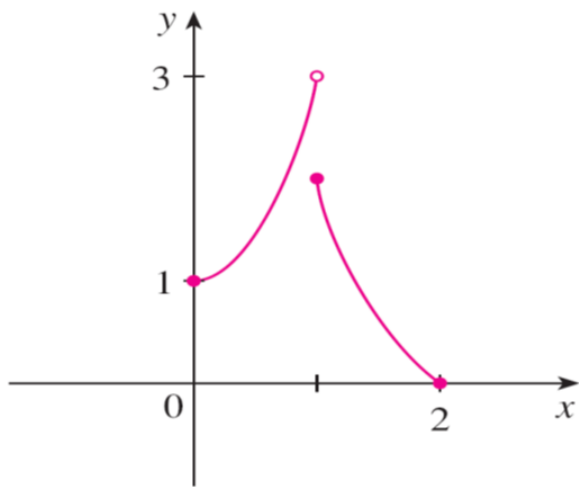
Functions continuous on a closed interval always attain extreme values.

Figure 8

Note that an extreme value can be taken on more than once.

Absolute and Local Extreme Values (9 of 11)

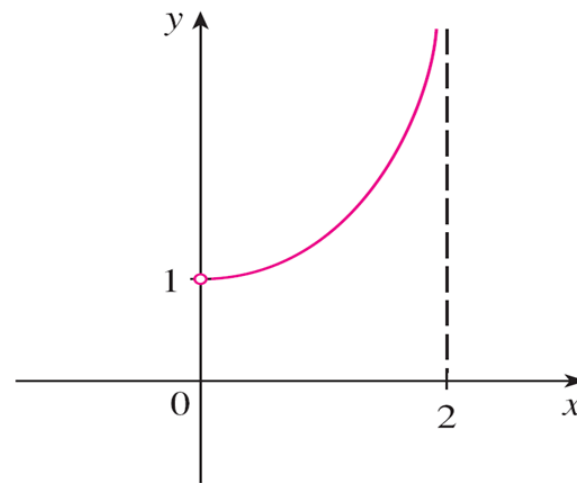
Figures 9 and 10 show that a function need not possess extreme values if either hypothesis (continuity or closed interval) is omitted from the Extreme Value Theorem.



This function has minimum value $f(2) = 0$, but no maximum value.

Figure 9

continuity



This continuous function g has no maximum or minimum.

Figure 10

closed interval

Absolute and Local Extreme Values (10 of 11)

The function f whose graph is shown in Figure 9 is defined on the closed interval $[0, 2]$ but has no maximum value. (Notice that the range of f is $[0, 3)$).

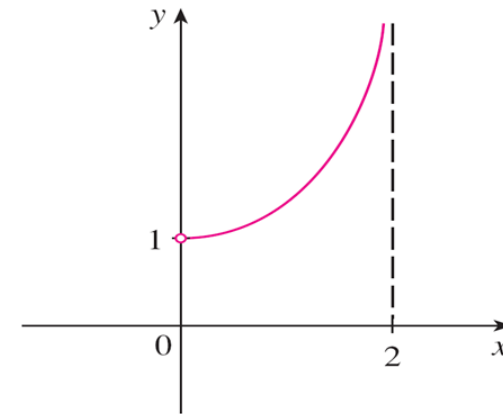
The function takes on values arbitrarily close to 3, but never actually attains the value 3.)

This does not contradict the Extreme Value Theorem because f is not continuous.

Absolute and Local Extreme Values (11 of 11)

The function g shown in Figure 10 is continuous on the open interval $(0, 2)$ but has neither a maximum nor a minimum value. [The range of g is $(1, \infty)$. The function takes on arbitrarily large values.]

This does not contradict the Extreme Value Theorem because the interval $(0, 2)$ is not closed.



This continuous function g has no maximum or minimum.

Figure 10

Critical Numbers and the Closed Interval Method

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Critical Numbers and the Closed Interval Method (1 of 5)

The Extreme Value Theorem says that a continuous function on a closed interval has a maximum value and a minimum value, but it does not tell us how to find these extreme values.

Figure 11 shows the graph of a function f with a local maximum at c and a local minimum at d .

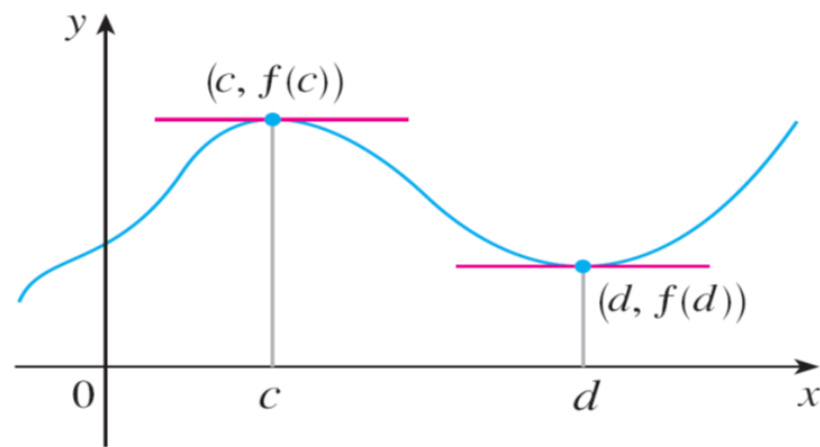


Figure 11

Critical Numbers and the Closed Interval Method (2 of 5)

It appears that at the maximum and minimum points the tangent lines are horizontal and therefore each has slope 0.

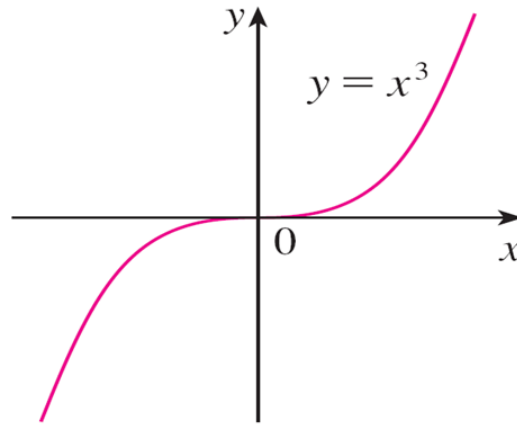
We know that the derivative is the slope of the tangent line, so it appears that $f'(c) = 0$ and $f'(d) = 0$. The following theorem says that this is always true for differentiable functions.

4 Fermat's Theorem If f has a local maximum or minimum at c , and if $f'(c)$ exists, then $f'(c) = 0$.

Example 5 (1 of 2)

If $f(x) = x^3$, then $f'(x) = 3x^2$, so $f'(0) = 0$.

But f has no maximum or minimum at 0, as you can see from its graph in Figure 12.



If $f(x) = x^3$, then $f'(0) = 0$ but f has no maximum or minimum.

Figure 12

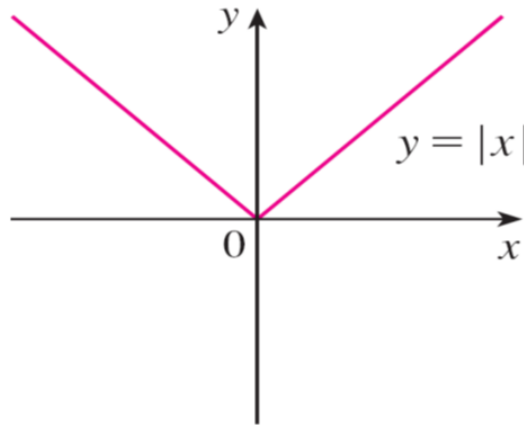
Example 5 (2 of 2)

The fact $f'(0) = 0$ simply means that the curve $y = x^3$ has a horizontal tangent at $(0, 0)$.

Instead of having a maximum or minimum at $(0, 0)$, the curve crosses its horizontal tangent there.

Example 6

The function $f(x) = |x|$ has its (local and absolute) minimum value at 0, but that value can't be found by setting $f'(x) = 0$ because, $f'(0)$ does not exist. (See Figure 13.)



If $f(x) = |x|$, then $f(0) = 0$ is a minimum value, but $f'(0)$ does not exist.

Figure 13

Critical Numbers and the Closed Interval Method (3 of 5)

Examples 5 and 6 show that we must be careful when using Fermat's Theorem. Example 5 demonstrates that even when $f'(c) = 0$ there need not be a maximum or minimum at c . (In other words, the converse of Fermat's Theorem is false in general.)

Furthermore, there may be an extreme value even when $f'(c)$ does not exist (as in Example 6).

Critical Numbers and the Closed Interval Method (4 of 5)

Fermat's Theorem does suggest that we should at least *start* looking for extreme values of f at the numbers c where $f'(c) = 0$ or where $f'(c)$ does not exist. Such numbers are given a special name.

6 Definition A **critical number** of a function f is a number c in the domain of f such that either $f'(c) = 0$ or $f'(c)$ does not exist.

In terms of critical numbers, Fermat's Theorem can be rephrased as follows.

7 If f has a local maximum or minimum at c , then c is a critical number of f .

Critical Numbers and the Closed Interval Method (5 of 5)

To find an absolute maximum or minimum of a continuous function on a closed interval, we note that either it is local or it occurs at an endpoint of the interval.

Thus the following three-step procedure always works.

The Closed Interval Method To find the *absolute* maximum and minimum values of a continuous function f on a closed interval $[a, b]$:

1. Find the values of f at the critical numbers of f in (a, b) .
2. Find the values of f at the endpoints of the interval.
3. The largest of the values from Steps 1 and 2 is the absolute maximum value; the smallest of these values is the absolute minimum value.



4.2

The Mean Value Theorem

The Mean Value Theorem (1 of 1)

We will see that many of the results depend on one central fact, which is called the Mean Value Theorem.



Rolle's Theorem

Rolle's Theorem (1 of 3)

To arrive at the Mean Value Theorem we first need the following result.

Rolle's Theorem Let f be a function that satisfies the following three hypotheses:

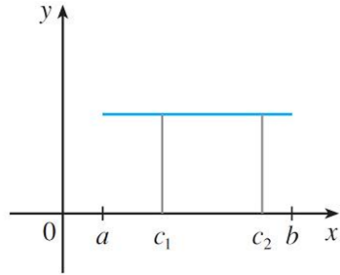
1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .
3. $f(a) = f(b)$

Then there is a number c in (a, b) such that $f'(c) = 0$.

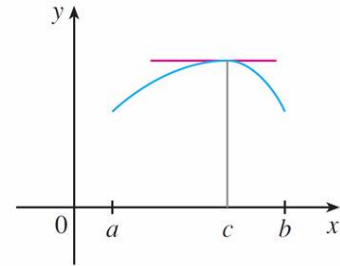
Before giving the proof let's take a look at the graphs of some typical functions that satisfy the three hypotheses.

Rolle's Theorem (2 of 3)

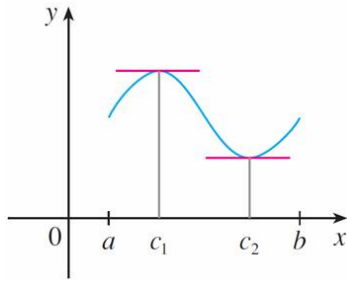
Figure 1 shows the graphs of four such functions.



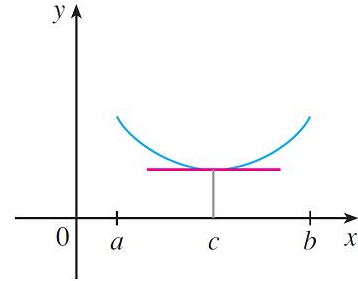
(a)



(b)



(c)



(d)

Figure 1

Rolle's Theorem (3 of 3)

In each case it appears that there is at least one point $(c, f(c))$ on the graph where the tangent is horizontal and therefore $f'(c) = 0$.

Thus Rolle's Theorem is plausible.

Example 2

Prove that the equation $x^3 + x - 1 = 0$ has exactly one real solution.

Solution:

First we use the Intermediate Value Theorem to show that a root exists.

Let $f(x) = x^3 + x - 1$. Then $f(0) = -1 < 0$ and $f(1) = 1 > 0$.

Since f is a polynomial, it is continuous, so the Intermediate Value Theorem states that there is a number c between 0 and 1 such that $f(c) = 0$.

Thus the given equation has a root.

Example 2 – Solution (1 of 2)

To show that the equation has no other real solution, we use Rolle's Theorem and argue by contradiction.

Suppose that it had two solutions a and b . Then $f(a) = 0 = f(b)$ and, since f is a polynomial, it is differentiable on (a, b) and continuous on $[a, b]$.

Thus, by Rolle's Theorem, there is a number c between a and b such that $f'(c) = 0$.

Example 2 – Solution (2 of 2)

But

$$f'(x) = 3x^2 + 1 \geq 1 \quad \text{for all } x$$

(since $x^2 \geq 0$) so $f'(x)$ can never be 0. This gives a contradiction.

Therefore the equation can't have two real roots.

The Mean Value Theorem

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The Mean Value Theorem (1 of 6)

Our main use of Rolle's Theorem is in proving the following important theorem, which was first stated by another French mathematician, Joseph-Louis Lagrange.

The Mean Value Theorem Let f be a function that satisfies the following hypotheses:

1. f is continuous on the closed interval $[a, b]$.
2. f is differentiable on the open interval (a, b) .

Then there is a number c in (a, b) such that

$$1 \quad f'(c) = \frac{f(b) - f(a)}{b - a}$$

or, equivalently,

$$2 \quad f(b) - f(a) = f'(c)(b - a)$$

The Mean Value Theorem (2 of 6)

Before proving this theorem, we can see that it is reasonable by interpreting it geometrically. Figures 3 and 4 show the points $A(a, f(a))$ and $B(b, f(b))$ on the graphs of two differentiable functions.

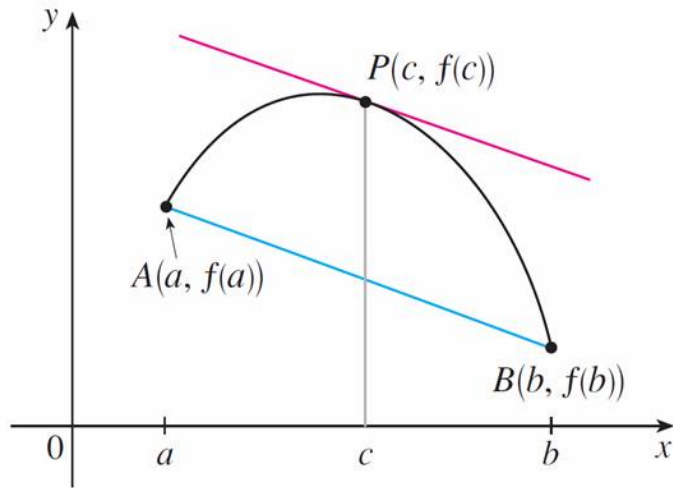


Figure 3

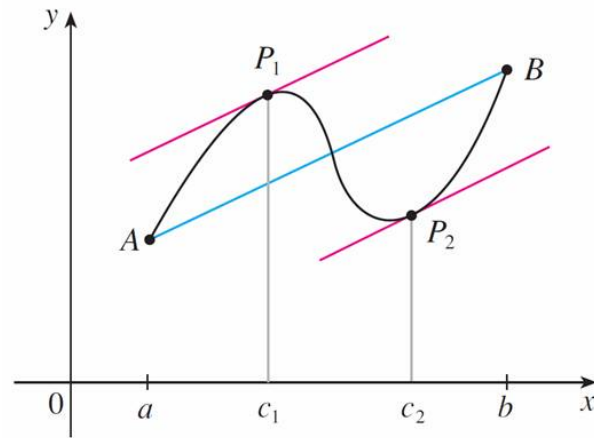


Figure 4

The Mean Value Theorem (3 of 6)

The slope of the secant line AB is

$$3 \quad m_{AB} = \frac{f(b) - f(a)}{b - a}$$

which is the same expression as on the right side of Equation 1.

The Mean Value Theorem (4 of 6)

Since $f'(c)$ is the slope of the tangent line at the point $(c, f(c))$, the Mean Value Theorem, in the form given by Equation 1, says that there is at least one point $P(c, f(c))$ on the graph where the slope of the tangent line is the same as the slope of the secant line AB .

In other words, there is a point P where the tangent line is parallel to the secant line AB .

Example 3 (1 of 3)

To illustrate the Mean Value Theorem with a specific function, let's consider

$$f(x) = x^3 - x, \quad a = 0, \quad b = 2.$$

Since f is a polynomial, it is continuous and differentiable for all x , so it is certainly continuous on $[0, 2]$ and differentiable on $(0, 2)$.

Therefore, by the Mean Value Theorem, there is a number c in $(0, 2)$ such that

$$f(2) - f(0) = f'(c)(2 - 0)$$

Example 3 (2 of 3)

Now $f(2) = 6$,

$f(0) = 0$, and

$f'(x) = 3x^2 - 1$, so this equation becomes

$$6 = (3c^2 - 1)2$$

$$= 6c^2 - 2$$

which
gives

$c^2 = \frac{4}{3}$, that is, $c = \pm \frac{2}{\sqrt{3}}$. But c must lie in $(0, 2)$, so $c = \frac{2}{\sqrt{3}}$.

Example 3 (3 of 3)

Figure 6 illustrates this calculation:

the tangent line at this value of c is parallel to the secant line OB .

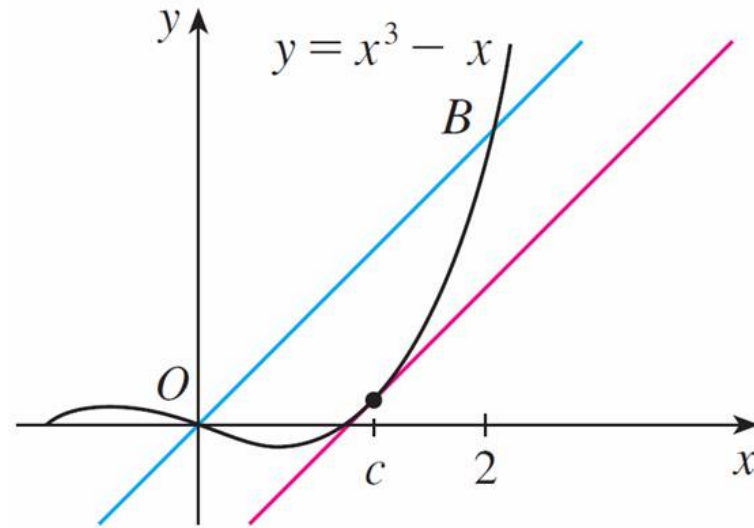


Figure 6

Example 5

Suppose that $f(0) = -3$ and $f'(x) \leq 5$ for all values of x . How large can $f(2)$ possibly be?

Solution:

We are given that f is differentiable (and therefore continuous) everywhere.

In particular, we can apply the Mean Value Theorem on the interval $[0, 2]$. There exists a number c such that

$$f(2) - f(0) = f'(c)(2 - 0)$$

Example 5 – Solution

So

$$f(2) = f(0) + 2f'(c) = -3 + 2f'(c)$$

We are given that $f'(x) \leq 5$ for all x , so in particular we know that $f'(c) \leq 5$.

Multiplying both sides of this inequality by 2, we have $2f'(c) \leq 10$, so

$$f(2) = -3 + 2f'(c) \leq -3 + 10 = 7$$

The largest possible value for $f(2)$ is 7.

The Mean Value Theorem (5 of 6)

The Mean Value Theorem can be used to establish some of the basic facts of differential calculus.

One of these basic facts is the following theorem.

5 Theorem If $f'(x) = 0$ for all x in an interval (a, b) , then f is constant on (a, b) .

7 Corollary If $f'(x) = g'(x)$ for all x in an interval (a, b) , then $f - g$ is constant on (a, b) ; that is, $f(x) = g(x) + c$ where c is a constant.

The Mean Value Theorem (6 of 6)

Note

Care must be taken in applying Theorem 5. Let

$$f(x) = \frac{x}{|x|} = \begin{cases} 1 & \text{if } x > 0 \\ -1 & \text{if } x < 0 \end{cases}$$

The domain of f is $D = \{x \mid x \neq 0\}$ and $f'(x) = 0$ for all x in D . But f is obviously not a constant function.

This does not contradict Theorem 5 because D is not an interval.



4.4

Indeterminate Forms and l'Hospital's Rule

Indeterminate Forms and l'Hospital's Rule (1 of 2)

Suppose we are trying to analyze the behavior of the function

$$F(x) = \frac{\ln x}{x - 1}$$

Although F is not defined when $x = 1$, we need to know how F behaves *near* 1. In particular, we would like to know the value of the limit

$$1 \quad \lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$$

Indeterminate Forms and l'Hospital's Rule (2 of 2)

In computing this limit we can't apply Law 5 of limits because the limit of the denominator is 0. In fact, although the limit in (1) exists, its value is not obvious because both numerator and denominator approach 0 and $\frac{0}{0}$ is not defined.



Indeterminate Forms $\left(\text{Types } \frac{0}{0}, \frac{\infty}{\infty} \right)$

Indeterminate Forms $\left(\text{Types } \frac{0}{0}, \frac{\infty}{\infty}\right)$ (1 of 5)

In general, if we have a limit of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where both $f(x) \rightarrow 0$ and $g(x) \rightarrow 0$ as $x \rightarrow a$, then this limit may or may not exist and is called an **indeterminate form of type** $\frac{0}{0}$.

Indeterminate Forms $\left(\text{Types } \frac{0}{0}, \frac{\infty}{\infty}\right)$ (2 of 5)

For rational functions, we can cancel common factors:

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{x^2 - x}{x^2 - 1} &= \lim_{x \rightarrow 1} \frac{x(x - 1)}{(x + 1)(x - 1)} \\ &= \lim_{x \rightarrow 1} \frac{x}{x + 1} \\ &= \frac{1}{2}\end{aligned}$$

We used a geometric argument to show that

$$\lim_{x \rightarrow 0} \frac{\sin x}{x} = 1$$

Indeterminate Forms $\left(\text{Types } \frac{0}{0}, \frac{\infty}{\infty}\right)$ (3 of 5)

But these methods do not work for limits such as (1).

Another situation in which a limit is not obvious occurs when we look for a horizontal asymptote of F and need to evaluate the limit

$$2 \quad \lim_{x \rightarrow \infty} \frac{\ln x}{x - 1}$$

It isn't obvious how to evaluate this limit because both numerator and denominator become large as $x \rightarrow \infty$.

Indeterminate Forms $\left(\text{Types } \frac{0}{0}, \frac{\infty}{\infty}\right)$ (4 of 5)

There is a struggle between numerator and denominator. If the numerator wins, the limit will be ∞ ; if the denominator wins, the answer will be 0. Or there may be some compromise, in which case the answer will be some finite positive number.

In general, if we have a limit of the form

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)}$$

where both $f(x) \rightarrow \infty$ (or $-\infty$) and $g(x) \rightarrow \infty$ (or $-\infty$), then the limit may or may not exist and is called an **indeterminate form of type** $\frac{\infty}{\infty}$.

Indeterminate Forms $\left(\text{Types } \frac{0}{0}, \frac{\infty}{\infty}\right)$ (5 of 5)

This type of limit can be evaluated for certain functions, including rational functions, by dividing numerator and denominator by the highest power of x that occurs in the denominator. For instance

$$\lim_{x \rightarrow \infty} \frac{x^2 - 1}{2x^2 + 1} = \lim_{x \rightarrow \infty} \frac{1 - \frac{1}{x^2}}{2 + \frac{1}{x^2}} = \frac{1 - 0}{2 + 0} = \frac{1}{2}$$

This method does not work for limits such as (2).



L'Hospital's Rule

L'Hospital's Rule (1 of 4)

L'Hospital's Rule Suppose f and g are differentiable and $g'(x) \neq 0$ on an open interval I that contains a (except possibly at a). Suppose that

$$\lim_{x \rightarrow a} f(x) = 0 \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = 0$$

or that

$$\lim_{x \rightarrow a} f(x) = \pm \infty \quad \text{and} \quad \lim_{x \rightarrow a} g(x) = \pm \infty$$

(In other words, we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.) Then

$$\lim_{x \rightarrow a} \frac{f(x)}{g(x)} = \lim_{x \rightarrow a} \frac{f'(x)}{g'(x)}$$

if the limit on the right side exists (or is ∞ or $-\infty$).

L'Hospital's Rule (2 of 4)

Note 1:

L'Hospital's Rule says that the limit of a quotient of functions is equal to the limit of the quotient of their derivatives, provided that the given conditions are satisfied. It is especially important to verify the conditions regarding the limits of f and g before using l'Hospital's Rule.

Note 2:

L'Hospital's Rule is also valid for one-sided limits and for limits at infinity or negative infinity; that is, " $x \rightarrow a$ " can be replaced by any of the symbols

$x \rightarrow a^+$, $x \rightarrow a^-$, $x \rightarrow \infty$, or $x \rightarrow -\infty$.

L'Hospital's Rule (3 of 4)

Note 3:

For the special case in which $f(a) = g(a) = 0$, f' and g' are continuous, and $g'(a) \neq 0$, it is easy to see why l'Hospital's Rule is true. In fact, using the alternative form of the definition of a derivative, we have

$$\begin{aligned}\lim_{x \rightarrow a} \frac{f'(x)}{g'(x)} &= \frac{f'(a)}{g'(a)} \\ &= \frac{\lim_{x \rightarrow a} \frac{f(x) - f(a)}{x - a}}{\lim_{x \rightarrow a} \frac{g(x) - g(a)}{x - a}}\end{aligned}$$

L'Hospital's Rule (4 of 4)

$$\begin{aligned} & \frac{f(x) - f(a)}{x - a} \\ = \lim_{x \rightarrow a} & \frac{x - a}{\frac{g(x) - g(a)}{x - a}} \\ & \frac{f(x) - f(a)}{g(x) - g(a)} = \lim_{x \rightarrow a} \frac{f(x)}{g(x)} \quad \left[\text{since } f(a) = g(a) = 0 \right] \end{aligned}$$

It is more difficult to prove the general version of l'Hospital's Rule.

Example 1

Find $\lim_{x \rightarrow 1} \frac{\ln x}{x - 1}$.

Solution:

Since

$$\lim_{x \rightarrow 1} \ln x = \ln 1 = 0 \quad \text{and} \quad \lim_{x \rightarrow 1} (x - 1) = 0$$

the limit is an indeterminate form of type $\frac{0}{0}$.

Example 1 – Solution

We can apply l'Hospital's Rule:

$$\begin{aligned}\lim_{x \rightarrow 1} \frac{\ln x}{x - 1} &= \lim_{x \rightarrow 1} \frac{\frac{d}{dx}(\ln x)}{\frac{d}{dx}(x - 1)} \\ &= \lim_{x \rightarrow 1} \frac{1}{x} \\ &= \lim_{x \rightarrow 1} \frac{1}{x} \\ &= 1\end{aligned}$$



Indeterminate Products (Type $0 \times \infty$)

Indeterminate Products (Type $0 \cdot \infty$) (1 of 3)

If $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = \infty$ (or $-\infty$), then it isn't clear what the value of $\lim_{x \rightarrow a} [f(x)g(x)]$, if any, will be. There is a struggle between f and g . If f wins, the answer will be 0; if g wins, the answer will be ∞ (or $-\infty$).

Or there may be a compromise where the answer is a finite nonzero number. This kind of limit is called an **indeterminate form of type** $0 \cdot \infty$.

Indeterminate Products (Type $0 \times \infty$) (2 of 3)

We can deal with it by writing the product fg as a quotient:

$$fg = \frac{f}{\frac{1}{g}} \quad \text{or} \quad fg = \frac{\frac{g}{1}}{f}$$

This converts the given limit into an indeterminate form of $\frac{0}{0}$ or $\frac{\infty}{\infty}$ so that we can use l'Hospital's Rule.

Example 6

Evaluate $\lim_{x \rightarrow 0^+} x \ln x$.

Solution:

The given limit is indeterminate because, as $x \rightarrow 0^+$, the first factor (x) approaches 0 while the second factor ($\ln x$) approaches $-\infty$.

Example 6 – Solution

Writing $x = \frac{1}{\left(\frac{1}{x}\right)}$, we have $\frac{1}{x} \rightarrow \infty$ as $x \rightarrow 0^+$, so l'Hospital's Rule gives

$$\begin{aligned}\lim_{x \rightarrow 0^+} x \ln x &= \lim_{x \rightarrow 0^+} \frac{\ln x}{\frac{1}{x}} \\&= \lim_{x \rightarrow 0^+} \frac{\frac{1}{x}}{-\frac{1}{x^2}} \\&= \lim_{x \rightarrow 0^+} (-x) \\&= 0\end{aligned}$$

Indeterminate Products (Type $0 \times \infty$) (3 of 3)

Note:

In solving Example 6 another possible option would have been to write

$$\lim_{x \rightarrow 0^+} x \ln x = \lim_{x \rightarrow 0^+} \frac{x}{\frac{1}{\ln x}}$$

This gives an indeterminate form of the type $\frac{0}{0}$, but if we apply l'Hospital's Rule we get a more complicated expression than the one we started with.

In general, when we rewrite an indeterminate product, we try to choose the option that leads to the simpler limit.



Indeterminate Differences (Type $\infty - \infty$)

Indeterminate Differences (Type $\infty - \infty$)

If $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = \infty$, then the
limit
$$\lim_{x \rightarrow a} [f(x) - g(x)]$$

is called an **indeterminate form of type $\infty - \infty$** . Again there is a contest between f and g .

Will the answer be ∞ (f wins) or will it be $-\infty$ (g wins) or will they compromise on a finite number? To find out, we try to convert the difference into a quotient (for instance, by using a common denominator, or rationalization, or factoring out a common factor) so that we have an indeterminate form of type $\frac{0}{0}$ or $\frac{\infty}{\infty}$.

Example 7

Compute $\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right)$.

Solution:

First notice that $\frac{1}{(\ln x)} \rightarrow \infty$ and $\frac{1}{(x-1)} \rightarrow \infty$ as $x \rightarrow 1^+$, so the limit is of type $\infty - \infty$. indeterminate

Here we can start with a common denominator :

$$\lim_{x \rightarrow 1^+} \left(\frac{1}{\ln x} - \frac{1}{x-1} \right) = \lim_{x \rightarrow 1^+} \frac{x-1-\ln x}{(x-1)\ln x}$$

Example 7 – Solution (1 of 2)

Both numerator and denominator have a limit of 0, so l'Hospital's Rule applies, giving

$$\begin{aligned}\lim_{x \rightarrow 1^+} \frac{x - 1 - \ln x}{(x - 1) \ln x} &= \lim_{x \rightarrow 1^+} \frac{1 - \frac{1}{x}}{(x - 1) \cdot \frac{1}{x} + \ln x} \\ &= \lim_{x \rightarrow 1^+} \frac{x - 1}{x - 1 + x \ln x}\end{aligned}$$

Again we have an indeterminate limit of type $\frac{0}{0}$.

Example 7 – Solution (2 of 2)

We apply l'Hospital's Rule a second time:

$$\begin{aligned}\lim_{x \rightarrow 1^+} \frac{x - 1}{x - 1 + x \ln x} &= \lim_{x \rightarrow 1^+} \frac{1}{1 + x \cdot \frac{1}{x} + \ln x} \\ &= \lim_{x \rightarrow 1^+} \frac{1}{2 + \ln x} = \frac{1}{2}\end{aligned}$$

Indeterminate Powers ($0^0, \infty^0, 1^\infty$)

Indeterminate Powers ($\text{Types } 0^0, \infty^0, 1^\infty$) (1 of 2)

Several indeterminate forms arise from the limit

$$\lim_{x \rightarrow a} [f(x)]^{g(x)}$$

1. $\lim_{x \rightarrow a} f(x) = 0$ and $\lim_{x \rightarrow a} g(x) = 0$ type 0^0
2. $\lim_{x \rightarrow a} f(x) = \infty$ and $\lim_{x \rightarrow a} g(x) = 0$ type ∞^0
3. $\lim_{x \rightarrow a} f(x) = 1$ and $\lim_{x \rightarrow a} g(x) = \pm \infty$ type 1^∞

Indeterminate Powers (Types $0^0, \infty^0, 1^\infty$) (1 of 2)

Each of these three cases can be treated either by taking the natural logarithm:

$$\text{let } y = [f(x)]^{g(x)}, \quad \text{then} \quad \ln y = g(x) \ln f(x)$$

or by writing the function as an exponential:

$$[f(x)]^{g(x)} = e^{g(x) \ln f(x)}$$

In either method we are led to the indeterminate product $g(x) \ln f(x)$, which is of type $0 \cdot \infty$.

Example 9

Calculate $\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x}$.

Solution:

First notice that as $x \rightarrow 0^+$, we have $1 + \sin 4x \rightarrow 1$ and $\cot x \rightarrow \infty$, so the given limit is (type 1^∞). Let indeterminate

$$y = (1 + \sin 4x)^{\cot x} \rightarrow \text{这里求导数要三乘!}$$

Then

$$\begin{aligned} \ln y &= \ln \left[(1 + \sin 4x)^{\cot x} \right] = \cot x \ln (1 + \sin 4x) \\ &= \frac{\ln (1 + \sin 4x)}{\tan x} \end{aligned}$$

Example 9 – Solution

So l'Hospital's Rule gives

$$\lim_{x \rightarrow 0^+} \ln y = \lim_{x \rightarrow 0^+} \frac{\ln(1 + \sin 4x)}{\tan x} = \lim_{x \rightarrow 0^+} \frac{\frac{4 \cos 4x}{1 + \sin 4x}}{\sec^2 x} = 4$$

So far we have computed the limit of $\ln y$, but what we want is the limit of y .

To find this we use the fact that $y = e^{\ln y}$:

$$\lim_{x \rightarrow 0^+} (1 + \sin 4x)^{\cot x} = \lim_{x \rightarrow 0^+} y = \lim_{x \rightarrow 0^+} e^{\ln y} = e^4$$



4.9

Antiderivatives

Antiderivatives (1 of 1)

<https://www.hku.hk>

A physicist who knows the velocity of a particle might wish to know its position at a given time.

An engineer who can measure the variable rate at which water is leaking from a tank wants to know the amount leaked over a certain time period.

A biologist who knows the rate at which a bacteria population is increasing might want to deduce what the size of the population will be at some future time.

In each case, the problem is to find a function whose derivative is a known function.



The Antiderivative of a Function

The Antiderivative of a Function (1 of 4)

If we have a function F whose derivative is the function f , then F is called an *antiderivative* of f .

Definition A function F is called an **antiderivative** of f on an interval I if $F'(x) = f(x)$ for all x in I .

The Antiderivative of a Function (2 of 4)

For instance, let $f(x) = x^2$. It isn't difficult to discover an antiderivative of f if we keep the Power Rule in mind. In fact, if $F(x) = \frac{1}{3}x^3$, then $F'(x) = x^2 = f(x)$.

But the function $G(x) = \frac{1}{3}x^3 + 100$ also satisfies $G'(x) = x^2$.

Therefore both F and G are antiderivatives of f .

Indeed, any function of the form $H(x) = \frac{1}{3}x^3 + C$, where C is a constant, is an antiderivative of f .

The Antiderivative of a Function (3 of 4)

The following theorem says that f has no other antiderivative.

1 Theorem If F is an antiderivative of f on an interval I , then the most general antiderivative of f on I is

$$F(x) + C$$

where C is an arbitrary constant.

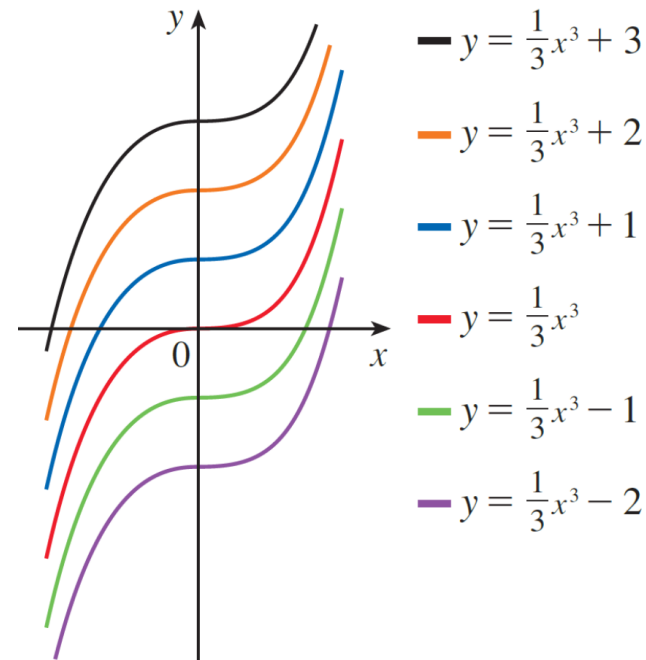
Going back to the function $\frac{1}{3}x^3 + C$.

$f(x) = x^2$, we see that the general antiderivative of f is

The Antiderivative of a Function (4 of 4)

By assigning specific values to the constant C , we obtain a family of functions whose graphs are vertical translates of one another (see Figure 1).

This makes sense because each curve must have the same slope at any given value of x .



Members of the family of antiderivatives of $f(x) = x^2$

Figure 1

Example 1

Find the most general antiderivative of each of the following functions.

(a) $f(x) = \sin x$

(b) $f(x) = \frac{1}{x}$

(c) $f(x) = x^n, n \neq -1$

Solution:

(a) If $F(x) = -\cos x$, then $F'(x) = \sin x$, so an antiderivative of $\sin x$ is $-\cos x$.

By Theorem 1, the most general antiderivative is $G(x) = -\cos x + C$.

Example 1 – Solution (1 of 3)

(b) We know that

$$\frac{d}{dx}(\ln x) = \frac{1}{x}$$

So on the interval $(0, \infty)$ the general antiderivative of $\frac{1}{x}$ is $\ln x + C$. We also learned that

$$\frac{d}{dx}(\ln |x|) = \frac{1}{x}$$

for all $x \neq 0$. Theorem 1 then tells us that the general antiderivative of $f(x) = \frac{1}{x}$ is $\ln |x| + C$ on any interval that doesn't contain 0. In particular, this is true on each of the intervals $(-\infty, 0)$ and $(0, \infty)$.

Example 1 – Solution (2 of 3)

So the general antiderivative of f is

$$F(x) = \begin{cases} \ln x + C_1 & \text{if } x > 0 \\ \ln(-x) + C_2 & \text{if } x < 0 \end{cases}$$

(c) We use the Power Rule to discover an antiderivative of x^n . In fact, if $n \neq -1$, then

$$\frac{d}{dx} \left(\frac{x^{n+1}}{n+1} \right) = \frac{(n+1)x^n}{n+1} = x^n$$

Therefore the general antiderivative of $f(x) = x^n$ is

$$F(x) = \frac{x^{n+1}}{n+1} + C$$

Example 1 – Solution (3 of 3)

This is valid for $n \geq 0$ since then $f(x) = x^n$ is defined on an interval. If n is negative (but $n \neq -1$), it is valid on any interval that doesn't contain 0.



Antidifferentiation Formulas

Antidifferentiation Formulas (1 of 3)

As in Example 1, every differentiation formula, when read from right to left, gives rise to an antidifferentiation formula. In Table 2 we list some particular antiderivatives.

2 Table of Antidifferentiation Formulas

Function	Particular antiderivative	Function	Particular antiderivative
$cf(x)$	$cf(x)$	$\sin x$	$\cos x$
$f(x) \pm g(x)$	$F(x) \pm G(x)$	$\sec^2 x$	$\tan x$
$x^n (n \neq -1)$	$\frac{x^{n+1}}{n+1}$	$\sec x \tan x$	$\sec x$
$\frac{1}{x}$	$\ln x $	$\frac{1}{\sqrt{1-x^2}}$	$\sin^{-1} x$
e^x	e^x	$\frac{1}{1+x^2}$	$\tan^{-1} x$
b^x	$\frac{b^x}{\ln b}$	$\cosh x$	$\sinh x$
$\cos x$	$\sin x$	$\sinh x$	$\cosh x$

To obtain the most general antiderivative from the particular ones in Table 2, we have to add a constant (or constants), as in Example 1.

Antidifferentiation Formulas (2 of 3)

Each formula in the table is true because the derivative of the function in the right column appears in the left column.

In particular, the first formula says that the antiderivative of a constant times a function is the constant times the antiderivative of the function.

The second formula says that the antiderivative of a sum is the sum of the antiderivatives. (We use the notation $F' = f, G' = g$.)

Example 2

Find all functions g such that $g'(x) = 4 \sin x + 2x^4 - x^{-\frac{1}{2}}$

$$g'(x) = 4 \sin x + \frac{2x^5 - \sqrt{x}}{x} \quad \begin{matrix} (x^4)' = (\frac{1}{5}x^5)' \\ (x^{-\frac{1}{2}})' = 2x^{\frac{1}{2}} \end{matrix}$$
$$g(x) = -4 \cos x + \frac{2}{5}x^5 - 2x^{\frac{1}{2}}$$

Solution:

We first rewrite the given function as follows:

$$g'(x) = 4 \sin x + \frac{2x^5}{x} - \frac{\sqrt{x}}{x} = 4 \sin x + 2x^4 - \frac{1}{\sqrt{x}}$$

Thus we want to find an antiderivative of

$$g'(x) = 4 \sin x + 2x^4 - x^{-1/2}$$

Example 2 – Solution

Using the formulas in Table 2 together with Theorem 1, we obtain

$$\begin{aligned} g(x) &= 4(-\cos x) + 2 \frac{x^5}{5} - \frac{x^{1/2}}{\frac{1}{2}} + C \\ &= -4 \cos x + \frac{2}{5} x^5 - 2\sqrt{x} + C \end{aligned}$$

Antidifferentiation Formulas (3 of 3)

An equation that involves the derivatives of a function is called a **differential equation**.

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The general solution of a differential equation involves an arbitrary constant (or constants) as in Example 2.

However, there may be some extra conditions given that will determine the constants and therefore uniquely specify the solution.



Graphing Antiderivatives

Graphing Antiderivatives (1 of 1)

If we are given the graph of a function f , it seems reasonable that we should be able to sketch the graph of an antiderivative F .

Suppose, for instance, that we are given that $F(0) = 1$. Then we have a place to start, the point $(0,1)$, and the direction in which we move our pencil is given at each stage by the derivative $F'(x) = f(x)$.

In the next example we use the principles of this chapter to show how to graph F even when we don't have a formula for f . This would be the case, for instance, when $f(x)$ is determined by experimental data.

Example 5

The graph of a function f is given in Figure 3. Make a rough sketch of an antiderivative F , given that $F(0) = 2$.

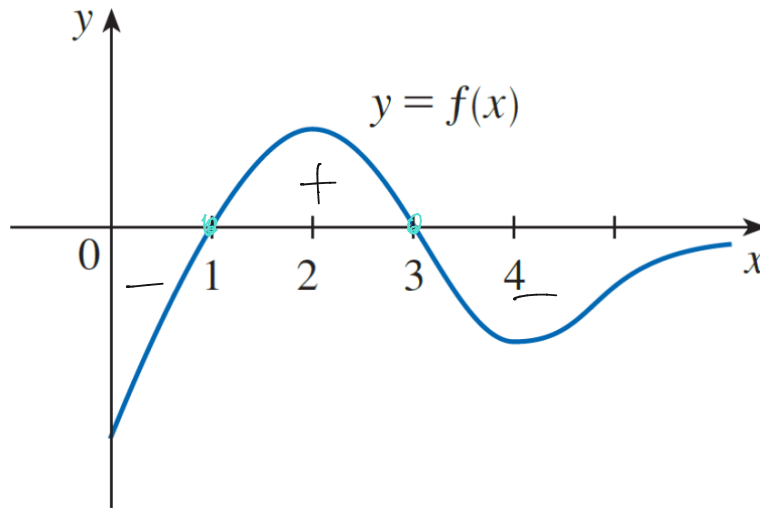


Figure 3

Example 5 – Solution (1 of 2)

We are guided by the fact that the slope of $y = F(x)$ is $f(x)$. We start at the point $(0, 2)$ and draw F as an initially decreasing function since $f(x)$ is negative when $0 < x < 1$.

Notice that $f(1) = f(3) = 0$, so F has horizontal tangents when $x = 1$ and $x = 3$. For $1 < x < 3$, $f(x)$ is positive and so F is increasing. We see that F has a local minimum when $x = 1$ and a local maximum when $x = 3$.

For $x > 3$, $f(x)$ is negative and so F is decreasing on $(3, \infty)$. Since $f(x) \rightarrow 0$ as $x \rightarrow \infty$, the graph of F becomes flatter as $x \rightarrow \infty$. Also notice that $F''(x) = f'(x)$ changes from positive to negative at $x = 2$ and from negative to positive at $x = 4$, so F has inflection points when $x = 2$ and $x = 4$.

Example 5 – Solution (2 of 2)

We use this information to sketch the graph of the antiderivative in Figure 4.

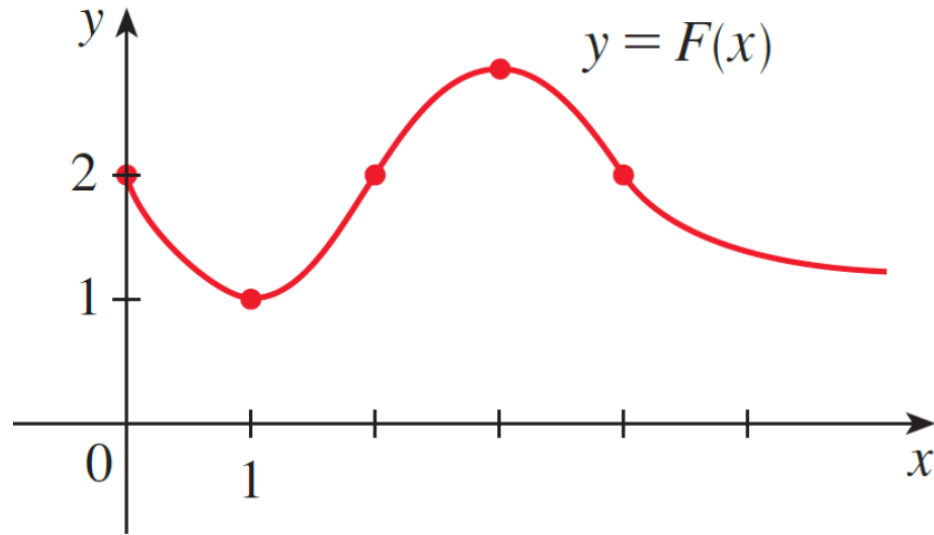


Figure 4



Linear Motion

Linear Motion (1 of 1)

Antidifferentiation is particularly useful in analyzing the motion of an object moving in a straight line. We know if the object has position function $s = f(t)$, then the velocity function $v(t) = s'(t)$.

is
This means that the position function is an antiderivative of the velocity function.

Likewise, the acceleration function $a(t) = v'(t)$, so the velocity function is an antiderivative of the acceleration.

If the acceleration and the initial values $s(0)$ and $v(0)$ are known, then the position function can be found by antidifferentiating twice.

Example 6

A particle moves in a straight line and has acceleration given by $a(t) = 6t + 4$. Its initial velocity is $v(0) = -6$ cm/s and its initial displacement is $s(0) = 9$ cm. Find its position function $s(t)$.

Solution:

Since $v'(t) = a(t) = 6t + 4$, antidifferentiation gives

$$\begin{aligned} v(t) &= 6 \frac{t^2}{2} + 4t + C \\ &= 3t^2 + 4t + C \end{aligned}$$

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$$s'(t) = v(t) \Rightarrow s(t) = -6t + 2t^2 + t^3 + b \Rightarrow b = 9$$

$$v'(t) = a(t) = 6t + 4 \Rightarrow v(t) = 3t^2 + 4t + a$$

$$v(0) = -6 \text{ cm/s} \Rightarrow a = -6$$

$$\therefore v(t) = 3t^2 + 4t - 6$$

Example 6 – Solution

Note that $v(0) = C$. But we are given that $v(0) = -6$, so $C = -6$ and

$$v(t) = 3t^2 + 4t - 6$$

Since $v(t) = s'(t)$, s is the antiderivative of v :

$$s(t) = 3 \frac{t^3}{3} + 4 \frac{t^2}{2} - 6t + D = t^3 + 2t^2 - 6t + D$$

This gives $s(0) = D$. We are given that $s(0) = 9$, so $D = 9$ and the required position function is

$$s(t) = t^3 + 2t^2 - 6t + 9$$